

## Final Exam Questions

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TTL: Quazi Shafayat Hossain

Please submit your questions by Saturday, 16 November 2024. Please make sure you make Set A and Set B of your question. Please provide a handwritten/latex solution with the marks distribution to your question.

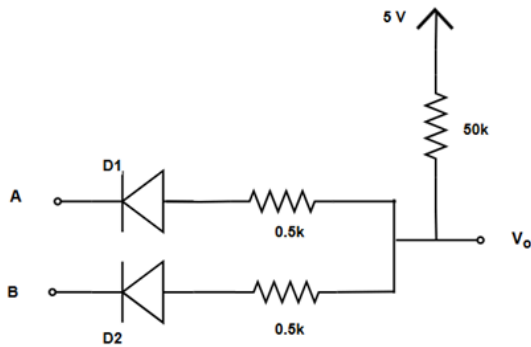
- While posting your question here, please follow this format:

|     |    |          |         |
|-----|----|----------|---------|
| (a) | CO | Question | [Marks] |
| (b) | CO | Question | [Marks] |
| (c) | CO | Question | [Marks] |

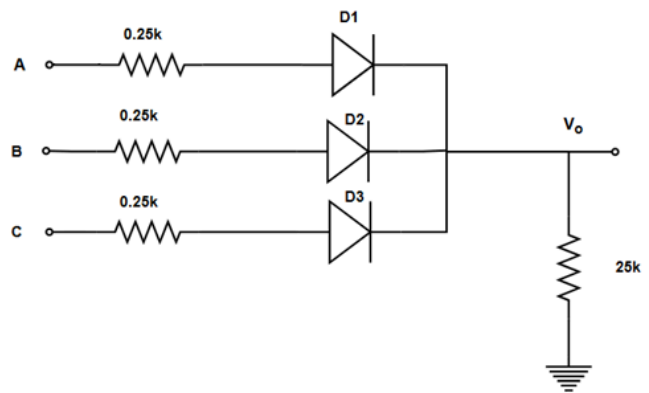
DL (SZZ, SDU):

### Set A

Answer the following questions from the given circuit. In this circuit input high logic and input low logic can be considered as 5 V and 0V. Diodes used in this circuit are Si diodes. You can use the Constant Voltage Drop model (conduction voltage 0.7 V) for analysis of diodes.



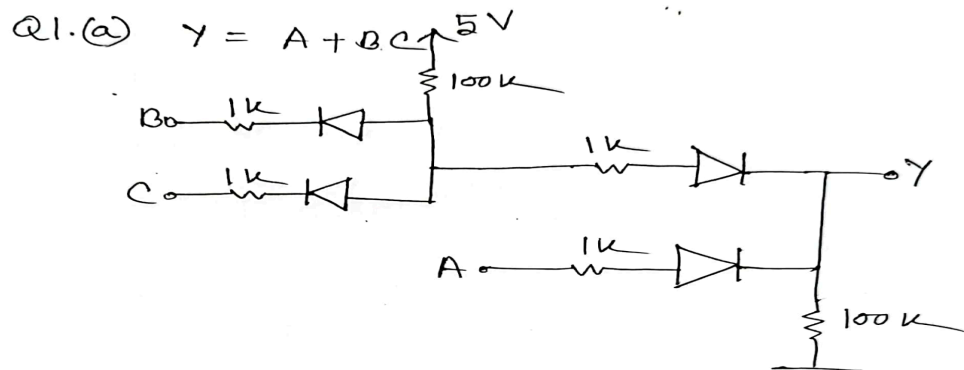
Q1 (b)



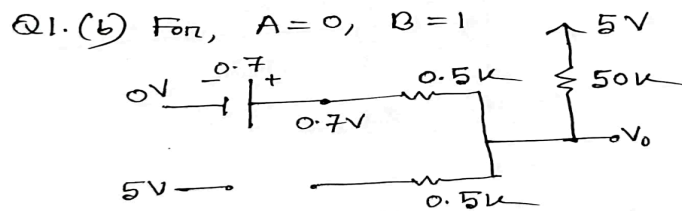
Q1 (c)

|     |     |                                                                                                                                                                |     |
|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| (a) | CO1 | Implement the Boolean expression ( $Y = A + BC$ ) using diode logic (DL) in terms of A, B and C.                                                               | [4] |
| (b) | CO1 | Find the output voltage of the circuit (Fig: Q1(b)) for the input combination ( $A = 0, B = 1$ ) and ( $A = 0, B = 0$ ).                                       | [8] |
| (c) | CO1 | Which combination of inputs will show maximum power dissipation for the circuit (Fig: Q1(c))? Calculate the value of maximum power for this input combination. | [8] |

Set A:

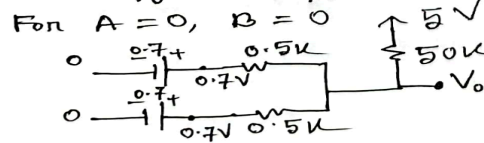


Resistors value can be used approximately.



$$\frac{V_o - 5}{50k} + \frac{V_o - 0.7}{0.5k} = 0$$

$$V_o = 0.743V$$



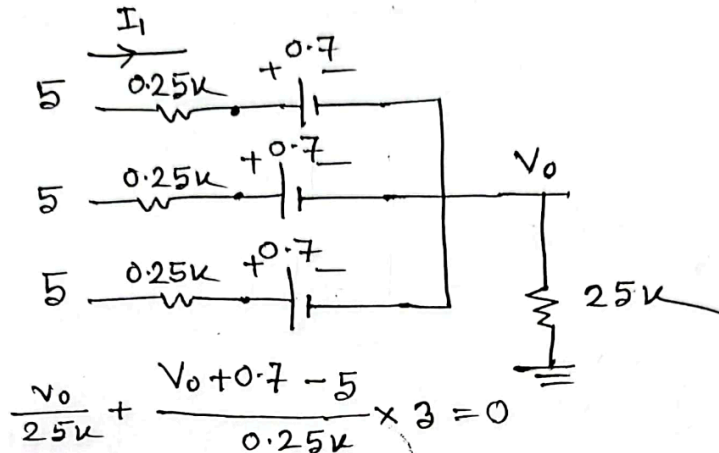
$$\frac{V_o - 5}{50k} + 2 \times \frac{V_o - 0.7}{0.5k} = 0$$

$$\Rightarrow V_o = 0.721V$$

Solve:

Q1.(c) For maximum power,

$$A=1, B=1, C=1$$



$$V_o = 4.2857 \text{ V}$$

$$P_{\text{max}} = (5 - 0) \times \frac{4.2857}{25k}$$

$$= 0.8571 \text{ mW}$$

$$\text{or } I_1 = \frac{5 - (V_o + 0.7)}{0.25k}$$

$$= \frac{5 - (4.2857 + 0.7)}{0.25k}$$

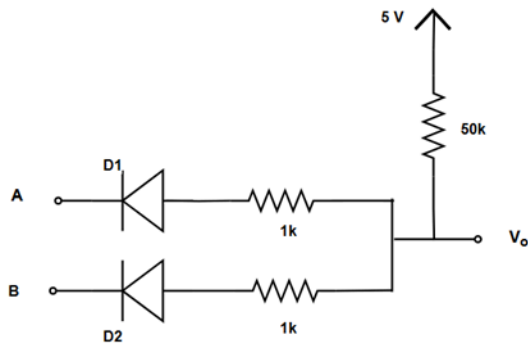
$$= 0.0572 \text{ mA}$$

$$P = 3 \times (5 - 0) \times 0.0572 \text{ mA}$$

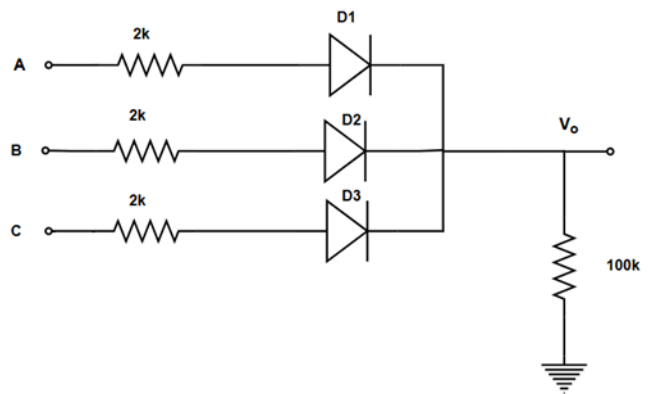
$$= 0.858 \text{ mW}$$

## Set B

Answer the following questions from the given circuit. In this circuit input high logic and input low logic can be considered as 5 V and 0V. Diodes used in this circuit are Si diodes. You can use the Constant Voltage Drop model (conduction voltage 0.7 V) for analysis of diodes.



Q1 (b)



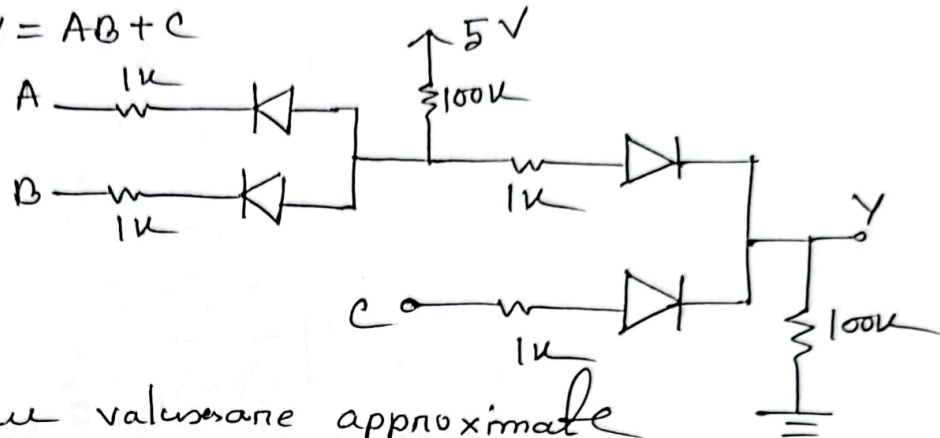
Q1 (c)

|     |     |                                                                                                                                                                |     |
|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| (a) | CO1 | Implement the Boolean expression ( $Y = AB + C$ ) using diode logic (DL) in terms of A, B and C.                                                               | [4] |
| (b) | CO1 | Find the output voltage of the circuit (Fig: Q1(b)) for the input combination (A = 1, B= 0) and (A = 0, B = 0).                                                | [8] |
| (c) | CO1 | Which combination of inputs will show maximum power dissipation for the circuit (Fig: Q1(c))? Calculate the value of maximum power for this input combination. | [8] |

**Solve:**

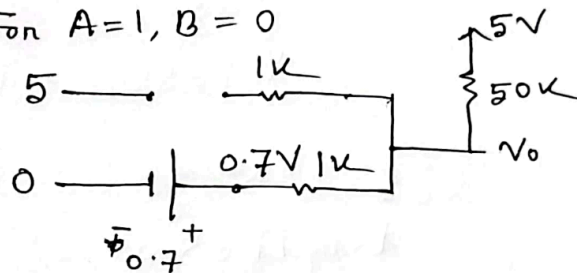
Set B

Q1. (a)  $Y = AB + C$



Resistor values are approximate

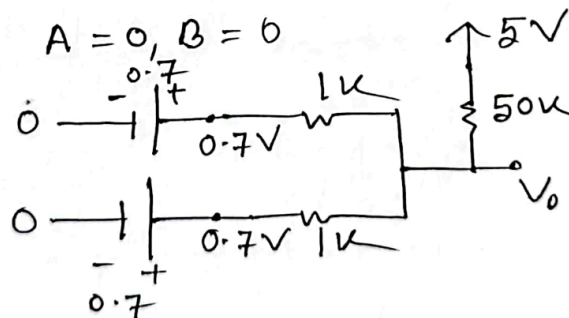
Q1. (b) For  $A=1, B=0$



$$\frac{V_0 - 5}{50k} + \frac{V_0 - 0.7}{1k} = 0$$

$$V_0 = 0.7842V$$

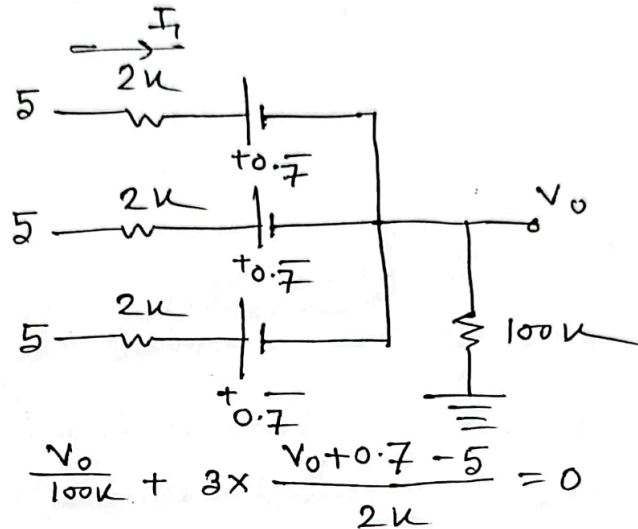
For  $A=0, B=0$



$$\frac{V_0 - 5}{50k} + 2 \times \frac{V_0 - 0.7}{1k} = 0$$

$$V_0 = 0.7426V$$

Q.1.(c) For maximum power dissipation,  
 $A = 1$ ,  $B = 1$ ,  $C = 1$



$$V_o = 4.2715V$$

$$P = (5 - 0) \times \frac{4.2715}{100k} = 0.2136mW$$

$$\text{or, } I_1 = \frac{5 - (V_o + 0.7)}{2k} = 0.01425mA$$

$$P = 3 \times (5 - 0) \times 0.01425mW$$

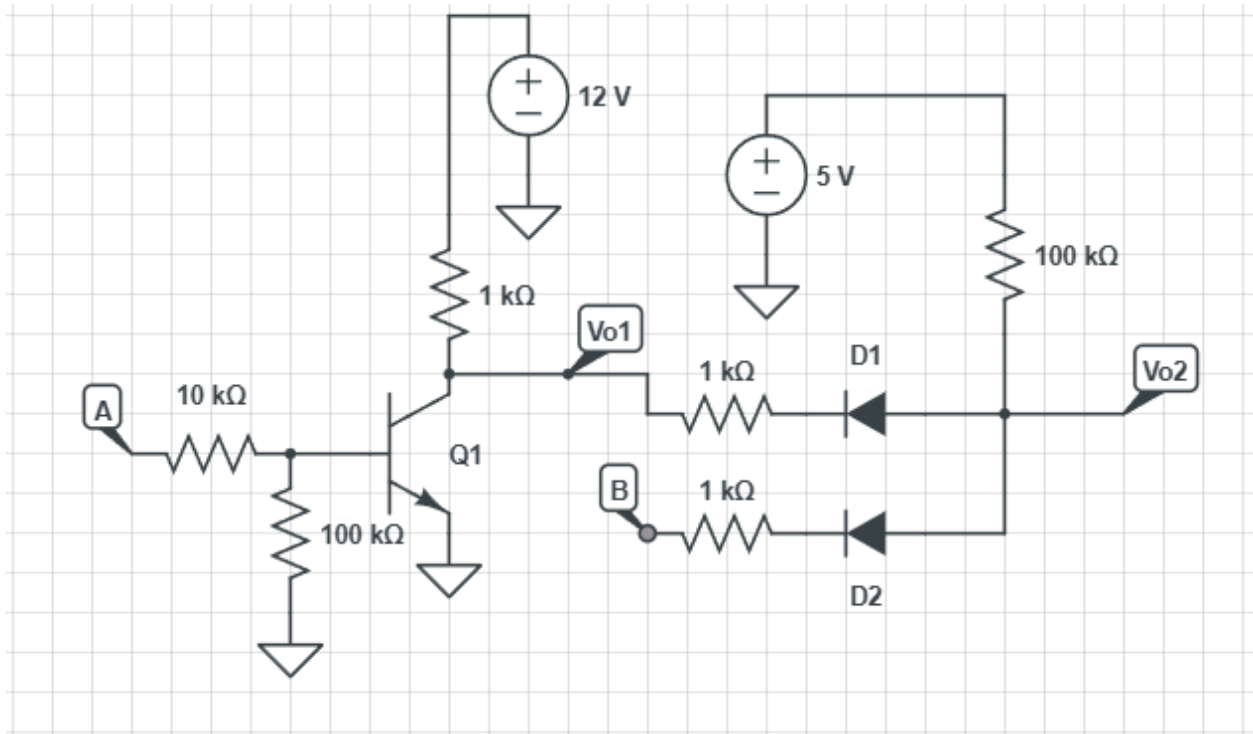
$$= 0.2137mW$$

## RTL (ZAZ, SDU):

### Set A

Answer the following questions from the given driver-load circuit. [ $V_{OL} = 0.2 \text{ V}$ ,

$V_{OH} = 11 \text{ V}$ ,  $\beta_F = 25$ ,  $\beta_{forced} = 0.1$ ] [High input = 12 V, Low input = 0.2 V]



|     |     |                                                                                     |     |
|-----|-----|-------------------------------------------------------------------------------------|-----|
| (a) | CO2 | Calculate the output voltage $V_{o2}$ for A = Logic 1, B = Logic 1.                 | [6] |
| (b) | CO2 | Calculate the power dissipation of the driver circuit for A = Logic 0, B = Logic 1. | [6] |
| (c) | CO2 | Calculate maximum fanout. [Consider B = Logic 1]                                    | [8] |

### Solve:

- (a) For A = 1, Q1 is in saturation, D2 off, D1 on.

$$V_{o1} = V_{OL} = 0.2$$

KCL at output,

$$\frac{5 - V_{o2}}{100k} = \frac{V_{o2} - 0.7 - V_{o1}}{1k}$$

$$\text{Solving, } V_{o2} = 0.94 \text{ V}$$

- (b) Transistor in cutoff, D1 off, D2 on

$$P_{(0)} = (0.2 - 0) * \frac{0.2 - 0}{100k + 10k} = 0.363 \mu W$$

(c) Case  $V_{o1} = 0.2V$  (Low):

D1 is on & current flows towards the collector from load. This current,

$$I_L = \frac{5 - 0.7 - 0.2}{101k} = 0.04 \text{ mA}$$

$$\text{No load collector current, } I'_C = \frac{12 - 0.2}{1k} = 11.8 \text{ mA}$$

Maximum fanout, N

$$\text{Then, } I_C = I'_C + I_L * N$$

Maximum allowable collector current

$$I_C = \beta_F * I_B = 25 * \left( \frac{12 - 0.8}{10k} - \frac{0.8}{100k} \right) = 27.8 \text{ mA}$$

$$\text{Solving, } N = \text{floor}(400) = 400$$

Case  $V_{o1} = 11V$  (High):

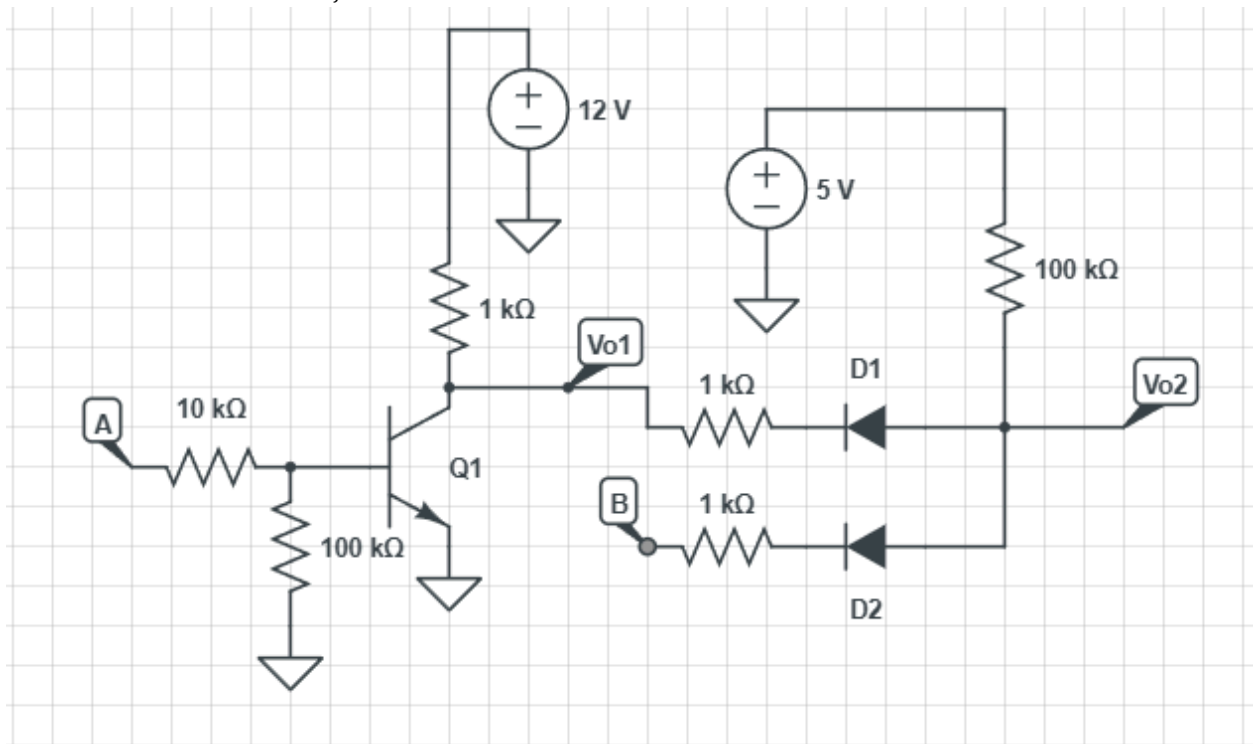
D1 is off, fanout is infinity

So, overall maximum fanout is 400.

### Set B

Answer the following questions from the given driver-load circuit. [ $V_{OL} = 0.2V$ ,

$V_{OH} = 10V$ ,  $\beta_F = 30$ ,  $\beta_{forced} = 0.2$ ] [High input = 12V, Low input = 0.2V]





|     |     |                                                                                     |     |
|-----|-----|-------------------------------------------------------------------------------------|-----|
| (a) | CO2 | Calculate the output voltage $V_{o2}$ for A = Logic 0, B = Logic 0.                 | [6] |
| (b) | CO2 | Calculate the power dissipation of the driver circuit for A = Logic 1, B = Logic 1. | [6] |
| (c) | CO2 | Calculate maximum fanout. [Consider B = Logic 1]                                    | [8] |

**Solve:**

- (a) For A = 0, Q1 in cutoff, D1 off, D2 on.

$$V_B = 0.2V$$

KCL at output,

$$\frac{5-V_{o2}}{100k} = \frac{V_{o2}-0.7-0.2}{1k}$$

$$\text{Solving, } V_{o2} = 0.94V$$

- (b) Transistor in saturation, D1 on, D2 off

$$P_{(1)} = (12 - 0) * \left( \frac{12-0.8}{10k} + \frac{12-0.2}{1k} \right) = 155.04 \text{ mW}$$

- (c) Case  $V_{o1} = 0.2V$  (Low):

D1 is on & current flows towards the collector from load. This current,

$$I_L = \frac{5-0.7-0.2}{101k} = 0.04 \text{ mA}$$

$$\text{No load collector current, } I'_C = \frac{12-0.2}{1k} = 11.8 \text{ mA}$$

Maximum fanout, N

$$\text{Then, } I_C = I'_C + I_L * N$$

Maximum allowable collector current

$$I_C = B_F * I_B = 30 * \left( \frac{12-0.8}{10k} - \frac{0.8}{100k} \right) = 33.36 \text{ mA}$$

$$\text{Solving, } N = \text{floor}(539) = 539$$

Case  $V_{o1} = 10V$  (High):

D1 is off, fanout is infinity

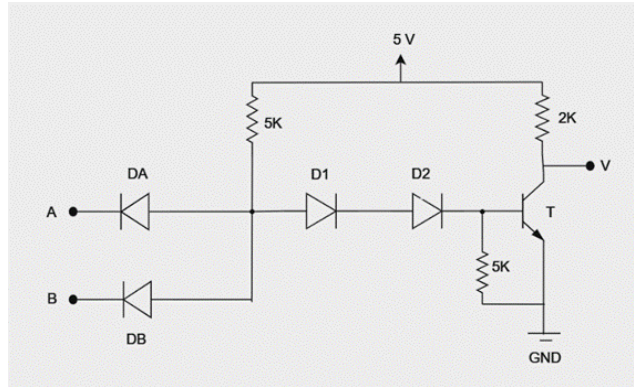
So, overall maximum fanout is 539.

## DTL (NUR):

**Set-A**

**20 marks**

Answer the following questions from the given circuit, where input low value is 0.2V and assume the input high . [  $V_{BE}(\text{cut in}) = 0.5V$ ,  $V_D(\text{cut in}) = 0.6V$ ,  $V_{BE}(\text{sat}) = 0.8V$ ,  $V_{CE}(\text{sat}) = 0.2V$ ,  $\beta_F = 30$  )

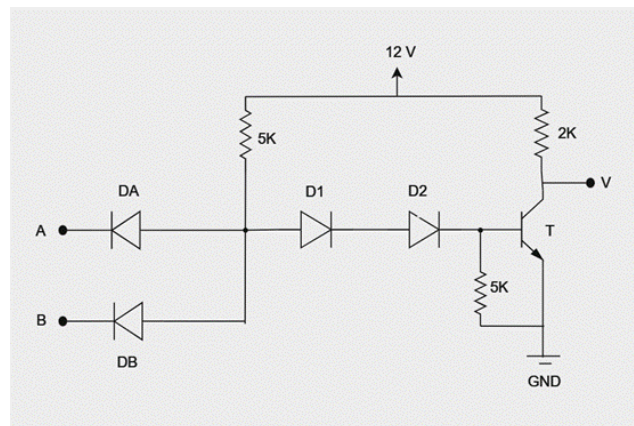


|     |     |                                                                                                                                   |       |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------|-------|
| (a) | CO2 | Create a truth table and determine which logic gate it represents, with justification. Also find $\beta_{\min}$ for this circuit. | [8+2] |
| (b) | CO2 | Find the average power dissipation.                                                                                               | [4]   |
| (c) | CO2 | Given that the loads are identical to the driver, determine the overall fanout.                                                   | [6]   |

### Set-B

20 marks

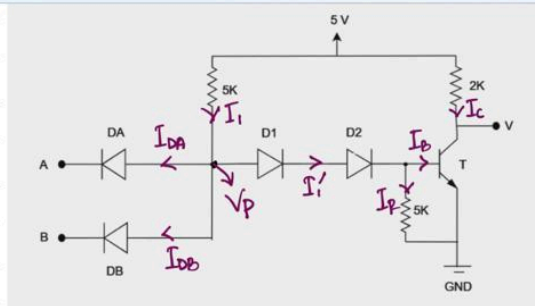
Answer the following questions from the given circuit, where input low value is 0.2V and assume the input high . [  $V_{BE}(\text{cut in}) = 0.5\text{V}$ ,  $V_D(\text{cut in}) = 0.6\text{V}$ ,  $V_{BE}(\text{sat}) = 0.8\text{V}$ ,  $V_{CE}(\text{sat}) = 0.2\text{V}$ ,  $\beta_F = 30$  )



|     |     |                                                                                                                                   |       |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------|-------|
| (a) | CO2 | Create a truth table and determine which logic gate it represents, with justification. Also find $\beta_{\min}$ for this circuit. | [8+2] |
| (b) | CO2 | Find the average power dissipation.                                                                                               | [4]   |

|     |     |                                                                                 |     |
|-----|-----|---------------------------------------------------------------------------------|-----|
| (c) | CO2 | Given that the loads are identical to the driver, determine the overall fanout. | [6] |
|-----|-----|---------------------------------------------------------------------------------|-----|

### Set-A Solution



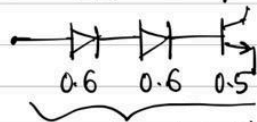
(a) Case-I

$$V_A = 0.2V, V_B = 0.2V$$

DA, DB → Conducting

$$\therefore 0.2V - 0.7V \rightarrow V_p = 0.9V$$

T → cutoff as  $V_p < 1.7V$



1.7V → min requirement for T to turn on.

$$I_1 = \frac{5-0.9}{5K} = 0.82mA, I_c = 0mA$$

$$I_c = \frac{5-V_o}{2} \Rightarrow [V_o = 5V]$$

# Case-IV:  $V_A = V_B = 5V$

DA, DB → off

T → Saturation.

$$V_{BE} = 0.8V \therefore V_p = 2.2V$$

$$V_{CE} = 0.2V \rightarrow [V_o = 0.2V]$$

$$I_1 = I_2 = \frac{5-2.2}{5K} = 0.56mA$$

$$I_c = \frac{5-0.2}{2K} = 2.4mA$$

$$I_R = \frac{0.8}{5K} = 0.16mA, I_B = I_1 - I_R = 0.4mA \rightarrow \beta_{min} = \frac{I_c}{I_B} = \frac{2.4}{0.4} = 6$$

Case-II

$$V_A = 0.2V, V_B = 5V$$

DA → Conducting, DB → off

$$V_p = 0.7 + 0.2 = 0.9V$$

T → cutoff →  $I_c = 0mA$

$$I_1 = \frac{5-0.9}{5K} = 0.82mA$$

$$[V_o = 5V]$$

Case-III

$$V_B = 0.2V, V_A = 5V$$

DB → Conducting, DA → off

$$V_p = 0.7 + 0.2 = 0.9V$$

T → cutoff →  $I_c = 0mA$

$$I_1 = \frac{5-0.9}{5K} = 0.82mA$$

$$[V_o = 5V]$$

| $V_A$ | $V_B$ | OTP V |
|-------|-------|-------|
| 0.2   | 0.2   | 5     |
| 0.2   | 5     | 5     |
| 5     | 0.2   | 5     |
| 5     | 5     | 0.2   |

∴ It's a NAND Gate.

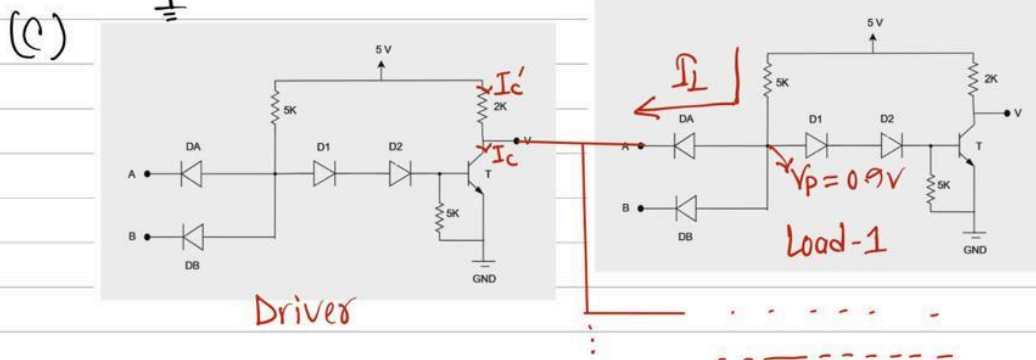
| (b) Case-I                                        | Case-II                                           | Case-III                                          |
|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| $\therefore P_1 = (5 - 0.2)I_1 = 3.93 \text{ mW}$ | $\therefore P_2 = (5 - 0.2)I_1 = 3.93 \text{ mW}$ | $\therefore P_3 = (5 - 0.2)I_1 = 3.93 \text{ mW}$ |

Case-IV

$$P_4 = (5 - 0)(I_1 + I_c) = 5 \times (0.56 + 2.4) = 14.8 \text{ mW}$$

$$\therefore \text{Avg } P_{\text{avg}} = \frac{P_1 + P_2 + P_3 + P_4}{4}$$

$P_{\text{avg}} = 6.64 \text{ mW}$



Case-I

Driver op  $\rightarrow V_0 = 5V$ ,  $D_A \rightarrow \text{off}$

$\therefore$  Demand current = 0, supply = 0

$\therefore \text{max fanout} = \infty$

Case-II : Driver op  $\rightarrow V_0 = 0.2V$ ,  $D_A \rightarrow \text{conducting}$ ,  $V_p = 0.9V$

$\therefore I_c' = \frac{5 - 0.2}{2} = 2.4 \text{ mA}$ ,  $I_L = \frac{5 - 0.9}{5K} = 0.82 \text{ mA}$

$N \rightarrow \text{fanout} \rightarrow \text{collector current}$ ,  $I_{c_{\text{max}}} = I_c' + N I_L$

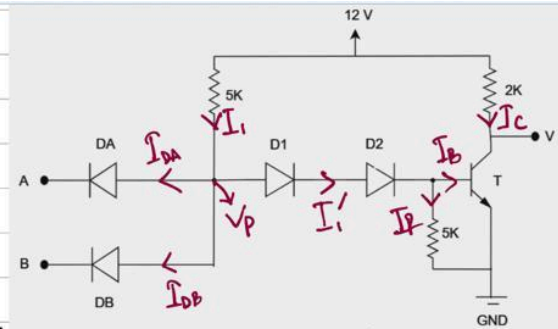
$\beta_{\text{forced}} \leq \frac{I_c}{I_B} \Rightarrow 12 = 2.4 + N \times 0.82$

$\beta_F = \frac{I_{c_{\text{max}}}}{I_B} \therefore N = 11.7 \approx 11$

$I_{c_{\text{max}}} = \beta_F I_B = 30 \times 0.4 = 12 \text{ mA}$

$\therefore \text{Overall fanout} = \min(\infty, 11) = 11 \text{ (Ans.)}$

# Set-B Solution



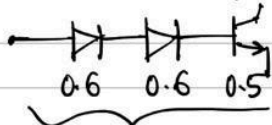
## Case-I

$$V_A = 0.2V, V_B = 0.2V$$

DA, DB  $\rightarrow$  Conducting

$$\therefore 0.2V \xrightarrow{0.7V} V_p = 0.9V$$

T  $\rightarrow$  Cutoff as  $V_p < 1.7V$



1.7V  $\rightarrow$  min requirement for T to turn on.

$$I_1 = \frac{12 - 0.9}{5K} = 2.22 \text{ mA}, I_c = 0 \text{ mA}$$

$$I_c = \frac{12 - V_o}{2} \Rightarrow V_o = 12$$

# Case-IV:  $V_A = V_B = 12V$

DA, DB  $\rightarrow$  off

T  $\rightarrow$  Saturation.

$$P \xrightarrow{0.7V} \xrightarrow{0.7V} V_{BE} = 0.8V \therefore V_p = 2.2V$$

$$V_{CE} = 0.2V \rightarrow V_o = 0.2V$$

$$I_1 = I_1' = \frac{12 - 2.2}{5K} = 1.96 \text{ mA}$$

$$I_c = \frac{12 - 0.2}{2K} = 5.9 \text{ mA}$$

$$I_R = \frac{0.8}{5K} = 0.16 \text{ mA}, I_B = I_1' - I_R = 1.8 \text{ mA} \rightarrow \beta_{min} = \frac{I_c}{I_B} = \frac{2.36}{1.8} = 1.31$$

## Case-II

$$V_A = 0.2V, V_B = 12V$$

DA  $\rightarrow$  Conducting, DB  $\rightarrow$  off

$$V_p = 0.7 + 0.2 = 0.9V$$

T  $\rightarrow$  cutoff  $\rightarrow I_c = 0 \text{ mA}$

$$I_1 = \frac{12 - 0.9}{5K} = 2.22 \text{ mA}$$

$$V_o = 12V$$

## Case-III

$$V_B = 0.2V, V_A = 12V$$

DB  $\rightarrow$  Conducting, DA  $\rightarrow$  off

$$V_p = 0.7 + 0.2 = 0.9V$$

T  $\rightarrow$  cutoff  $\rightarrow I_c = 0 \text{ mA}$

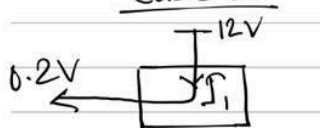
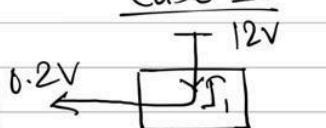
$$I_1 = \frac{12 - 0.9}{5K} = 2.22 \text{ mA}$$

$$V_o = 12V$$

| $V_A$ | $V_B$ | OTP V |
|-------|-------|-------|
| 0.2   | 0.2   | 0.2   |
| 0.2   | 12    | 12    |
| 12    | 0.2   | 12    |
| 12    | 12    | 0.2   |

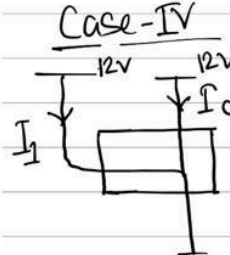
$\therefore$  It's a NAND Gate.



| Case-I                                                                            | Case-II                                                                           | Case-III                                                                           |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |  |
| $\therefore P_1 = (12 - 0.2)I_1$<br>$= 26.196 \text{ mW}$                         | $\therefore P_2 = (12 - 0.2)I_1$<br>$= 26.196 \text{ mW}$                         | $\therefore P_3 = (12 - 0.2)I_1$<br>$= 26.196 \text{ mW}$                          |

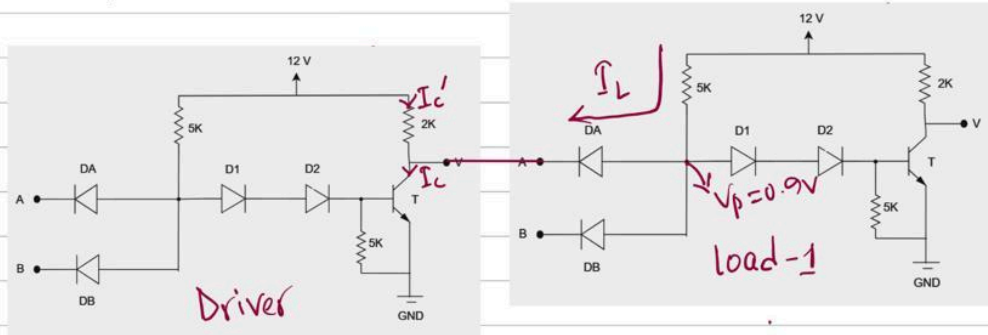
Case-IV



$P_4 = (12 - 0)(I_1 + I_c)$   
 $= 12 \times (1.96 + 5.9)$   
 $= 94.32 \text{ mW}$

$\therefore \text{Avg } P_{\text{avg}} = \frac{P_1 + P_2 + P_3 + P_4}{4}$   

$P_{\text{avg}} = 43.22 \text{ mW}$



Case-I

Driver op  $\rightarrow V_0 = 12 \text{ V}$ ,  $D_A \rightarrow \text{off}$

$\therefore$  Demand current = 0, supply = 0

$\therefore$  max fanout =  $\infty$

Case-II : Driver op  $\rightarrow V_0 = 0.2 \text{ V}$ ,  $D_A \rightarrow \text{conducting}$ ,  $V_p = 0.9 \text{ V}$

$\therefore I_c' = \frac{12 - 0.2}{2} = 5.9 \text{ mA}$ ,  $I_L = \frac{12 - 0.9}{5K} = 2.22 \text{ mA}$

$N \rightarrow \text{fanout} \rightarrow \text{collector current}$ ,  $I_{c_{\text{max}}} = I_c' + N I_L$

$P_{\text{forced}} \leq \frac{I_c}{I_B}$

$P_F = \frac{I_{c_{\text{max}}}}{I_B}$

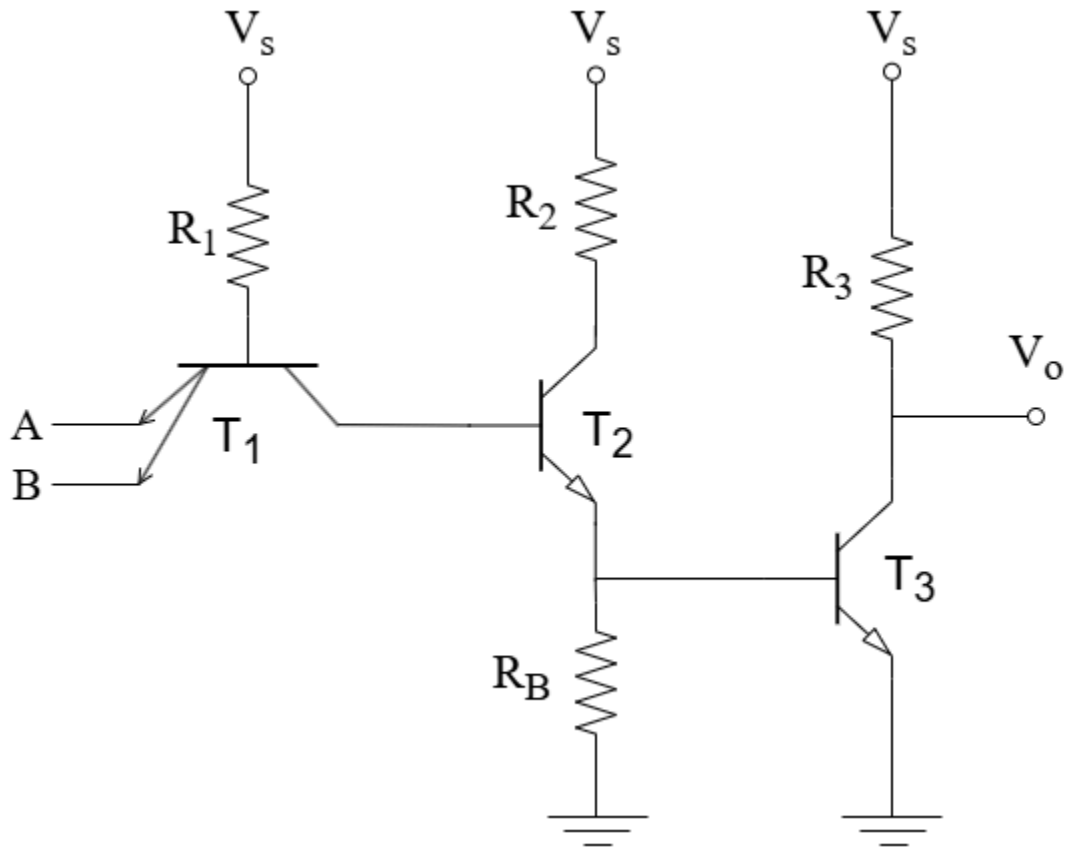
$I_{c_{\text{max}}} = P_F I_B = 30 \times 1.8 = 54 \text{ mA}$

$\Rightarrow 54 = 5.9 + N \times 2.22$

$\therefore N = 21.67 \approx 21$

$\therefore \text{Overall fanout} = \min(\infty, 21)$   
 $= 21 \text{ (Ans.)}$

## TTL (QSH):



**Set A:**  $V_s = 9\text{ V}$ ,  $R_2 = 2\text{ k}$ ,  $R_3 = 4.4\text{ k}$ ,  $R_B = 4.4\text{ k}$

**Set B:**  $V_s = 11\text{ V}$ ,  $R_2 = 2.5\text{ k}$ ,  $R_3 = 5.4\text{ k}$ ,  $R_B = 4.4\text{ k}$

|     |     |                                                                                                                                                                                                                                                                             |    |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| (a) | CO1 | Determine the power dissipation of this TTL circuit when both inputs are HIGH. You must show your calculation. Assume $R_1 = 6.7\text{ k}/8.7\text{ k}$ for this question only                                                                                              | 14 |
| (b) | CO1 | Suppose you are selling this TTL logic gate to a customer and he states a strict criteria of keeping the power dissipation below $1.23\text{ mW}/1.87\text{ mW}$ when both inputs are LOW (0 V). Determine the maximum value of $R_1$ you can use to satisfy this criteria. | 6  |

### Solution:

(For both sets, most calculations converge to the same result)

$$(a) I_{B1} = \frac{V_S - 2.3}{R_1} = 1 \text{ mA}$$

$$I_{E1a} = I_{E1b} = \beta_R * I_{B1} = 0.1 * 1 = 0.1 \text{ mA}$$

$$I_{C1} = I_{B2} = I_{B1} + I_{E1a} + I_{E1b} = 1.2 \text{ mA}$$

$$I_{C2} = \frac{V_S - 1}{R_2} = 4 \text{ mA}$$

$$I_{E2} = I_{C2} + I_{B2} = 4 + 1.2 = 5.2 \text{ mA}$$

$$I_{RB} = \frac{0.8}{R_B} = 0.2 \text{ mA}$$

$$I_{B3} = I_{E2} - I_{RB} = 5 \text{ mA}$$

$$I_{C3} = \frac{V_S - 0.2}{R_3} = 2 \text{ mA}$$

$$I_{E3} = I_{C3} + I_{B3} = 2 + 5 = 7 \text{ mA}$$

### Power Dissipation

$$= \sum \Delta V * I$$

$$= (V_S - 0) (I_{RB} + I_{E3})$$

$$\Rightarrow \mathbf{64.8 \text{ mW}} \text{ for Set A} \quad | \quad \mathbf{79.2 \text{ mW}} \text{ for Set B}$$

$$(b) \text{Power Dissipation} = \sum \Delta V * I$$

$$PD = (V_S - 0) * \frac{V_S - 0.8}{R_1}$$

$$R_1 = \frac{V_S * (V_S - 0.8)}{PD}$$

$$R_1 = \mathbf{60 \text{ k}\Omega}$$